

# Khan Ong Fernandez

co230532@adzu.edu.ph | 0994-900-7934 | Ayala, Zamboanga City

## OBJECTIVE

To apply my system development skills and embedded systems knowledge toward building secure, mission-critical systems and automating workflows while expanding my expertise into cybersecurity and defense-specific technology.

## EDUCATION

|                                                                                                                               |              |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Ateneo de Zamboanga University</b> <ul style="list-style-type: none"><li>Bachelor of Science in Computer Science</li></ul> | 2023-Present |
|-------------------------------------------------------------------------------------------------------------------------------|--------------|

## AFFILIATIONS

|                                                                                                    |              |
|----------------------------------------------------------------------------------------------------|--------------|
| <b>ADZU Reserved Officer Training Corps (ROTC)</b><br>Current Cadet Lieutenant Colonel   2nd Class | 2024-Present |
| <b>Ateneo Informatics Computing Guild</b><br>Member                                                | 2024-2024    |
| <b>ADZUSPLOIT</b><br>Member                                                                        | 2024-Present |
| <b>Ateneo Fiesta Marketing and Promotion Committee</b><br>Videographer and Editor                  | 2024-Present |

## EXPERIENCES

- Developed an embedded automation system for soil moisture and temperature regulation using DHT11 and soil moisture sensors, enabling real-time environmental monitoring and control.
- Developed a modular traffic control system utilizing linked lists and finite state machines to simulate real-time traffic flow and pedestrian management logic.
- Assisted in backend debugging and API integration by analyzing data-fetching issues and resolving UI rendering inconsistencies.
- Conducted research on lightweight preprocessing techniques on facial recognition accuracy under constrained hardware and variable environmental conditions.
- Developed a localized disaster response app integrating the Open-Meteo API for real-time weather monitoring, a WebView interface for official "Alerto Zamboanga" warnings, and a dedicated safety module with interactive emergency guidelines and local hotlines.

## SKILLS

**Development:** C++, HTML, CSS, JavaScript, Java, XML, React, Laravel, SQL, PostgreSQL, Embedded Systems, Prototyping, UI Design  
**Creative & Tools:** Videography, Video Editing (DaVinci Resolve), Figma, Arduino IDE, VS Code, XAMPP, Android Studio

## CERTIFICATIONS

|                                                            |      |
|------------------------------------------------------------|------|
| PAF Basic ROTC   Ateneo de Zamboanga University            | 2024 |
| PAF Summer Cadre Training   Batch Sinag Animo              | 2024 |
| PAF Summer Cadre Training   Volunteer Reservist Instructor | 2025 |
| Hack The Box University CTF 2025                           | 2025 |
| Ateneo Fiesta 2025   Marketing and Promotion Committee     | 2025 |